

Integer word moments do not control fractional Schatten norms: a counterexample to Open Problem 1 of Alhejji–Knill

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Abstract

Alhejji and Knill ask whether word-trace dominance between two pairs of positive semidefinite matrices with matched spectra forces domination of the Schatten p -norms of their sums for all $p \in [1, \infty)$. A positive answer would extend the spin-alignment conjecture from integer Schatten norms to all Schatten norms and to the von Neumann entropy. We answer in the negative. The simplest witness is a commuting, diagonal family in dimension three, for which word dominance is analytic and the p -norm inequality fails at $p = 3/2$ in exact integers. The failure is therefore classical: non-commutativity plays no role in it.

1. The problem

Let A_0, A_1, B_0, B_1 be positive semidefinite with $\lambda(A_0) = \lambda(B_0)$ and $\lambda(A_1) = \lambda(B_1)$, and for a binary word s write $\Pi_s(A) = \prod_i A_{s(i)}$. Open Problem 1 of [1] asks: if $\text{tr } \Pi_s(A) \geq |\text{tr } \Pi_s(B)|$ for every s , must $\|A_0 + A_1\|_p \geq \|B_0 + B_1\|_p$ for every $p \in [1, \infty)$? It is posed as a question, so an instance satisfying the hypothesis and violating the conclusion resolves it.

2. A commuting counterexample

Take

$$A_0 = \text{diag}(176, 0, 64), \quad A_1 = \text{diag}(80, 49, 0), \quad B_0 = \text{diag}(64, 176, 0), \quad B_1 = A_1.$$

Spectra match by inspection and all four are positive semidefinite. Because everything commutes, each word collapses to a product of powers, and dominance reduces to the single analytic statement

$$(64/176)^k + (49/80)^m \leq \frac{859}{880} < 1 \quad (k, m \geq 1),$$

so the hypothesis holds for all words. The conclusion fails at $p = 3/2$: the two spectral sums $\sum \lambda^{3/2}$ are 4951 and 5103, so $\|A_0 + A_1\|_{3/2} < \|B_0 + B_1\|_{3/2}$. The comparison is between integers; there is no numerical tolerance involved.

The extremal member of the family, $\text{diag}(2, 0, 1), \text{diag}(2, 1, 0)$ against $\text{diag}(1, 2, 0), \text{diag}(2, 1, 0)$, has failure window exactly $(1, 2)$: the function $f(p) = 4^p + 2 - 2 \cdot 3^p$ has zeros exactly at $p = 1$ and $p = 2$ by Descartes' rule.

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3. A non-commuting instance, and minimality

A 3×3 non-commuting instance with $\lambda(A_0) = \lambda(B_0) = \{0, 1/5, 7/10\}$ and $\lambda(A_1) = \lambda(B_1) = \{1/10, 3/10, 1/2\}$ was found first. It has equality at $p = 1$ and fails on the whole interval $p \in (1, p^*)$ with $p^* = 1.97584\dots$, the endpoints pinned by a Descartes/Laguerre sign argument. Word dominance for this instance is not analytic; it rests on a cone argument for a family of 2×2 transfer blocks, with a resonance constant at which the inequality saturates asymptotically from below without crossing. A separate argument rules out $d = 2$, so qutrits are minimal.

4. Verification and what is missing

All load-bearing steps are exact rational or 100-digit arithmetic. The 3×3 instance was re-derived from scratch by an independent agent with its own frame and cone argument, reaching the same resonance constant; hostile search covered every exact word to length 13 with no violation. The diagonal family was found by a container-isolated run with no network access and no sight of the earlier work.

What is missing is a human-typeset, referee-grade proof of the all-lengths word-dominance step for the 3×3 instance — every scalar inequality in it was discharged symbolically, so this is a write-up obligation and not a gap. For the diagonal family the kernel lemma is corroborated on thousands of draws but not yet proved symbolically.

Disclosure

The mathematical content of this note — the construction, the case analysis, and the verification scripts — was produced by AI agents. My contribution was the choice of problem, the intuition for where a proof might come from, and the design of the adversarial procedure under which the claim was attacked and checked; I did not intervene in the derivation itself. Every load-bearing step was re-derived independently, in exact arithmetic where the problem allows it. This note is a record of a computer-assisted result. It has not been refereed.

References

- [1] M. A. Alhejji and E. Knill, *Towards the spin-alignment conjecture*, arXiv:2307.06894.
- [2] Z. Song and L. Chen, *A counterexample to the strong spin alignment conjecture*, arXiv:2603.25410.